Data Dictionary

Version 2.1.0: 13 Nov 2026

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This document defines the variables contained in the four CSV files deposited in the repository (https://github.com/miltheo/SCOPE-MOVE) in folder /analysis/inputs. Definitions and extraction rules follow the project extraction manual (https://osf.io/bk7yr/files/yk8a3), with additional derived variables used for harmonised reporting and meta-analysis in the accompanying study (future citation).

Missing data codes

Code	Meaning
n.r.	Not reported in the source study.
n.a.	Not applicable for the study/model.
None	Explicitly not done / not used (as stated by authors).
Unclear	Could not be determined after full-text review and checking cited sources.

Where these codes appear as literal strings in a column, treat them as missingness indicators rather than valid values.

Files and relational structure

1) Study_Characteristics.csv: one row per included study (study-level metadata). Primary key: id.

2) Extraction_Classifiers.csv: one row per classifier validation (validation-level). Primary key: val_id, with numeric study identifier id for joining to Study_Characteristics.id.

3) Extraction_EnergyExpenditure.csv: one row per energy-expenditure validation (validation-level). Primary key: val_id, with numeric study identifier id for joining to Study_Characteristics.id. Four empty CSV records are excluded from analyses.

4) Quality_Assessment.csv: QUADAS-2 signalling questions and domain judgements, one row per study. Joins to Study_Characteristics.id using Covidence #; Study ID is the author-year display label.

Extraction_Classifiers.csv

Rows: 245. Columns: 38.

Column	Type	Description	Allowed/Example values	Notes
val_id	integer	Unique validation identifier (validation-level row id).		Primary key for validation-level tables.
id	integer	Internal study identifier used across extracted tables (study-level).		Joins Study_Characteristics.id to validation-level tables.
study	text	Study label in the format First author (year).		Corresponds to 'Study ID' in the extraction manual. Use numeric id for study-level joins.
Is the model published previously?	text	Whether the evaluated model was previously published before the validation study.		
movement_behaviour	categorical	Movement behaviour domain assessed by the prediction model.	Activity intensity; Activity type; Sleep	Sleep denotes sleep-wake classification.
model_type	text	High-level model type (e.g., multi-class classifier).	Binary classifier; Multi-class classifier; Multiple binary classifiers	
model_test_outcomes	text	Outcome classes or continuous outcome specification used for testing.		
model_test_classification	text	Detailed definitions of the tested outcome classes, including thresholds or decision boundaries where specified.	Sleep vs wake; sedentary vs non-sedentary; n.a.	
test_criterion	text	Criterion/reference method used for validation.	DO; DO; VO; IC; PSG; VO	Common codes: DO direct observation, VO video observation, PSG polysomnography, IC indirect calorimetry, DLW doubly labelled water.
Reported?	binary	Reporting flag recorded in the extraction sheet.	1; 0; blank	Retained extraction flag. Analytic selection uses the metric values, not this flag.
CM?	text	Indicator that a confusion matrix with counts was available for classifiers.	%; EC; NO	% for percentage reporting, EC for event counts; NO when not reported

age_group	categorical	Age group category for the cohort used for model development/testing.	Children and Youth; Adults; Older adults; n.r.	Children and Youth: 3-17 yrs; Adults: 18-65 yrs; Older adults: >65 yrs.
test_device	text	Wrist device used in the validation.		
test_model	text	Specific prediction model/method name as reported.		
health	categorical	Participant health status category.	Healthy; Clinical; Mixed; Wheelchair users; n.r.	Study-level category used for summaries.
wear_site	text	Wear site for the device (e.g., non-dominant wrist).		
features	text	Types of features input in the prediction model.	Time-domain; Frequency-domain; Spatial; Time-Frequency; Non-linear	
single_feature	text	Specific model input feature(s), transformations and units recorded during extraction.	ENMO (mg); MAD (mg); feature lists; n.a.; n.r.	May contain more than one feature; the source column name is retained.
method	categorical	Method family used for grouping (classifiers table).	Threshold / Cutpoint; Heuristic Rule; Linear / GLM; Classical ML; Deep Learning; Probabilistic Temporal; Hybrid / Ensemble	
method_sub	text	More specific method label within the method family.		
train_environment	text	Environment where training data were collected.	Free-living; Hybrid; Laboratory	
train_protocol	text	Training protocol structure.	Hybrid; Structured; Unstructured	
test_environment	text	Environment where testing/validation data were collected.	Free-living; Hybrid; Laboratory	
test_protocol	text	Testing protocol structure.	Hybrid; Structured; Unstructured	
validation_type	text	Validation approach used for evaluation.	Apparent; Cross-validation; External; Hold-out	
f1_mean	numeric	Macro-averaged F1 score (percentage, 0-100).		
f1_sd	numeric	Standard deviation or uncertainty for F1 (percentage points).		

sens_mean	numeric	Sensitivity (true positive rate) (percentage, 0-100).		
sens_sd	numeric	Standard deviation or uncertainty for sensitivity (percentage points).		
spec_mean	numeric	Specificity (true negative rate) (percentage, 0-100).		
spec_sd	numeric	Standard deviation or uncertainty for specificity (percentage points).		
metric_origin	categorical	How class-based metrics were averaged/derived.	macro; micro; per_class; aggregate_cm; unclear; n.a.	
sd_origin	categorical	Source of SD/SE for the metric.	cv_folds; classes; participants; cv+classes; external_test; not_reported; n.a.	
k_folds	text	Number of folds/replicates in the outer validation loop.		Stored as text in CSV; numeric when available.
n_classes	text	Number of outcome classes over which performance was computed.	10; 2; 3; 4; 5; 6; n.a.	
n_participants	text	Number of unique participants contributing to the validation/test set.		Stored as text to preserve n.a./n.r. codes.
other	text	Free-text flags and notes used during extraction and harmonisation.		
doi	text	Digital Object Identifier or persistent link for the study.		

Extraction_EnergyExpenditure.csv

Rows: 79 CSV records (75 validation records and 4 empty records). Columns: 31.

Column	Type	Description	Allowed/Example values	Notes
val_id	integer	Unique validation identifier (validation-level row id).		Primary key for validation-level tables.
id	integer	Internal study identifier used across extracted tables (study-level).		Joins Study_Characteristics.id to validation-level tables.
study	text	Study label in the format First author (year).		Corresponds to 'Study ID' in the extraction manual. Use numeric id for study-level joins.
Is the model published previously?	text	Whether the evaluated model was previously published before the validation study.	Yes; No -Hildebrand 2014	
movement_behaviour	categorical	Movement behaviour domain assessed by the prediction model.	Energy expenditure	
model_type	text	High-level model type (e.g., multi-class classifier).	Supervised regression	
model_test_outcomes	text	Outcome classes or continuous outcome specification used for testing.		
model_test_classification	text	Detailed definitions of the tested outcome classes, including thresholds or decision boundaries where specified.	Sleep vs wake; sedentary vs non-sedentary; n.a.	Not applicable to the continuous energy-expenditure outcome.
test_criterion	text	Criterion/reference method used for validation.	DLW; IC; VO	Common codes: DO direct observation, VO video observation, PSG polysomnography, IC indirect calorimetry, DLW doubly labelled water.
age_group	categorical	Age group category for the cohort used for model development/testing.	Children and Youth; Adults; Older adults; n.r.	Children and Youth: 3-17 yrs; Adults: 18-65 yrs; Older adults: >65 yrs.
test_device	text	Wrist device used in the validation.		
health	categorical	Participant health status category.	Healthy; Clinical; Mixed; Wheelchair users; n.r.	Study-level category used for summaries. Healthy is apparently healthy.

wear_site	text	Wear site for the device (e.g., non-dominant wrist).	Both wrists; ; Dominant wrist; Left wrist; Non-dominant wrist; Right wrist; n.r.	
features	text	Types of features input in the prediction model.		
single_feature	text	Specific model input feature(s), transformations and units recorded during extraction.	ENMO (mg); MAD (mg); feature lists; n.a.; n.r.	May contain more than one feature; the source column name is retained.
method	categorical	Method family used for grouping (classifiers table).	Learning; Probabilistic Temporal; Hybrid / Ensemble; Non-linear / Mixed Statistical	
method_sub	text	More specific method label within the method family.		
method_group	categorical	Binary grouping of methods (energy expenditure table).	Threshold/Heuristic or Linear Models; Machine Learning or Other Models	
train_environment	text	Environment where training data were collected.	Free-living; Hybrid; Laboratory	
train_protocol	text	Training protocol structure.	Hybrid; Structured; Unstructured	
test_environment	text	Environment where testing/validation data were collected.	Free-living; Hybrid; Laboratory	
test_protocol	text	Testing protocol structure.	Hybrid; Structured; Unstructured	
validation_type	text	Validation approach used for evaluation.	Apparent; Cross-validation; External; Hold-out	
mape_mean	numeric	Mean absolute percentage error for energy expenditure estimation (percentage).		
mape_sd	numeric	Standard deviation or uncertainty for MAPE (percentage points).	12.97; 14; 15.1; 4.57; 5.1; 5.26; 7.89; 9.49; 9.52; n.r.	
rmse_met_mean	numeric	Root mean square error in METs, if reported.		
rmse_met_sd	numeric	Standard deviation or uncertainty for RMSE in METs.	0.028; 0.059; 0.06; 0.13; 0.21; 0.36; 0.67; 1.7; n.r.	
n_participants	text	Number of unique participants contributing to the validation/test set.	10; 15; 17; 24; 30; 33; 36; 51; 59; 7; 9; n.a.	Stored as text to preserve n.a./n.r. codes.

sd_origin	categorical	Source of SD/SE for the metric.	cv_folds; classes; participants; cv+classes; external_test; not_reported; n.a.	
k_folds	text	Number of folds/replicates in the outer validation loop.	10; 15; 17; 30; 59; 7; 9; n.a.	Stored as text in CSV; numeric when available.
other	text	Free-text flags and notes used during extraction and harmonisation.		

Quality_Assessment.csv

Rows: 155. Columns: 18.

See https://osf.io/bk7yr/files/a8sq7 for the Quality Assessment tool or refer to the original study.

Column	Type	Description	Allowed/Example values	Notes
Covidence #	integer	Covidence record identifier for the included study.		Joins Study_Characteristics.id.
Study ID	text	Study label (First author year) for QUADAS-2 assessment.		
Title	text	Study title in the assessment sheet.		
SQ1	categorical	QUADAS-2 signalling question 1 response.	Low risk of bias; High risk of bias	
SQ2	categorical	QUADAS-2 signalling question 2 response.	Low risk of bias; High risk of bias	
SQ3	categorical	QUADAS-2 signalling question 3 response.	Low risk of bias; High risk of bias	
SQ4	categorical	QUADAS-2 signalling question 4 response.	Low risk of bias; High risk of bias	
Domain 1: Patient Selection/Study Design	categorical	Overall risk of bias judgement for QUADAS-2 Domain 1.	Low risk of bias; High risk of bias	
SQ5	categorical	QUADAS-2 signalling question 5 response.	Low risk of bias; High risk of bias	
SQ6	categorical	QUADAS-2 signalling question 6 response.	Low risk of bias; High risk of bias	
Domain 2: Index Measure	categorical	Overall risk of bias judgement for QUADAS-2 Domain 2.	Low risk of bias; High risk of bias	
SQ7	categorical	QUADAS-2 signalling question 7 response.	Low risk of bias; High risk of bias	
SQ8	categorical	QUADAS-2 signalling question 8 response.	Low risk of bias; High risk of bias	
Domain 3: Criterion Measure	categorical	Overall risk of bias judgement for QUADAS-2 Domain 3.	Low risk of bias; High risk of bias	
SQ9	categorical	QUADAS-2 signalling question 9 response.	Low risk of bias; High risk of bias	
SQ10	categorical	QUADAS-2 signalling question 10 response.	Low risk of bias; High risk of bias	
SQ11	categorical	QUADAS-2 signalling question 11 response.	Low risk of bias; High risk of bias	
Domain 4: Flow and Timing	categorical	Overall risk of bias judgement for QUADAS-2 Domain 4.	Low risk of bias; High risk of bias	

Study_Characteristics.csv

Rows: 155. Columns: 31.

Column	Type	Description	Allowed/Example values	Notes
id	integer	Internal study identifier used across extracted tables (study-level).		Joins Study_Characteristics.id to validation-level tables.
study	text	Study label in the format First author (year).		Corresponds to 'Study ID' in the extraction manual. Use numeric id for study-level joins.
year	integer	Publication year of the study.		
country	text	Countries where the study was conducted (semicolon-separated if multiple).		
age_group	categorical	Age group category for the cohort used for model development/testing .	Children and Youth; Adults; Older adults; n.r.	Children and Youth: 3-17 yrs; Adults: 18-65 yrs; Older adults: >65 yrs.
sample_size	integer	Total recruited sample size for the study (study-level).		
females	integer	Number of female participants (if reported).		If not reported, coded as n.r. at extraction time.
health	categorical	Participant health status category.	Healthy; Clinical; Mixed; Wheelchair users; n.r.	Study-level category used for summaries.
ethnicity_reported	categorical	Whether ethnicity distribution was reported.	Yes; 1; No; n.r.	
unique_model	text	Indicator that the study contributed a unique prediction model.		
device	text	Wearable device model as reported (study-level).		
device_brand	text	Wearable device brand.		
sensor	text	Accelerometer sensor chip/model where reported.	ADXL330 (Analog Devices); ADXL335 (Analog Device); BMA150 (Bosch Sensortec); KXSC7-3672 (Kionix); n.r.	
sensor_type	categorical	Accelerometer sensor type.	Uniaxial; Biaxial; Triaxial; Other; n.r.	

dynami_range	text	Full-scale/dynamic range of accelerometer (e.g., ±8 g).	±6g; ±16g; ±5g; n.r.; ±1.5g; ±16g; ±2g; ±3g; ±4g; ±5g; ±6g; ±6g; ±8g; ±8g; ±8g; ±16g; ±8g	Column header spelling retained from source file.
bit_resolution	text	Bit resolution of the accelerometer output (e.g., 12-bit) or other reported resolution.	10 bit; 12 bit; 13 bit; 16 bit; 4 mg; 8 bit; n.r.	
acquisition_sampling_rate	text	Native sampling rate used for data collection (Hz).		
sampling_rate_cat	categorical	Categorised sampling rate for reporting.	1-20 Hz; 21-50 Hz; 51-80 Hz; 81-100 Hz; >100 Hz; Variable; n.r.	
resampled	categorical	Indicator that raw data were resampled prior to feature extraction.	Yes; 1 No; n.r.	
resampled_sampling_rate	text	Target sampling rate after resampling (Hz), if applicable.		
raw_signal_check	categorical	Whether calibration/quality control checks on raw signals were reported.	Yes; 1 No; n.r.	
digital_filter	categorical	Whether a digital filter was applied (as reported).	Yes; 1 No; None; n.r.	
digital_filter_type	text	Filter type (e.g., low-pass, band-pass), if applied.	Band-pass; High-pass; Low-pass; n.r.	
highpass_cutoff	text	High-pass cutoff frequency (Hz), if applicable.		
lowpass_cutoff	text	Low-pass cutoff frequency (Hz), if applicable.		
gravity_removal	text	Approach used to treat/remove gravitational component.	De-trend subtraction; FIlter; Subtracting 1g from VM; None; Subtracting mean; Subtracting prior 5 sec moving average; n.a.	
feature_types	text	Feature domains extracted (study-level).		Semicolon-separated list.
window_size	text	Window length/epoch used for feature extraction.		

method_best	text	Best-performing method (study-level summary).	Classical ML; Deep Learning; Heuristic Rule; Hybrid / Ensemble; Linear / GLM; Non-linear / Mixed Statistical; Probabilistic Temporal; Threshold / Cutpoint	
sub_method_best	text	Best-performing sub-method (study-level summary).		
outcomes	text	Movement behaviour outcomes assessed in the study (semicolon-separated).	Activity intensity; Activity type; Energy expenditure; Sleep-Wake	